Course's Name: BDA
Course's Number: 20626-101
Questions' Number: 3 Questions
Total Mark: 30
Time: 50 Minutes

UCAS

Instructor's Name: Dr. Osama Hamed
Student's Name: ____________
Student's Number: ___________
Section's Number: ___________

Midterm Exam 2nd Semester 2025/26

Question #1: (7.5 Points)

1) You are working as Data Scientist on a BDA project; you are also responsible for the guidance of trainees. Explain in a simple language the necessary steps required to create a Python development project on the trainee's laptop. Your answer should be written in human language (not GenAI) and constructed based on the explanations provided in the recordings. **(15 M)**

Scientific Python لانو ما بنحملها تقليدي ولا بنعملها extract فبتوفر وقت
وفيها مكتبات جاهزة
البيئة الافتراضية من خلال python -m ven r (name file or folder)
من مجلدر scripts بنفعل امر activities فاسم البيئة بيظهر قدام
المؤشروو بهيك اشتغلت
بنثبت ال modules زي pandas , sql alchemy , y data protifiling
قبلهم بنحط pip install
Ps بنعمل كذا مجلد زي جيت هب data , src , scripts
بنعمل batch وبنحط فيه الاوامر اللي بدنا نشغل عليها ide اتوماتيكي
بنربطه مع github وبنعمل push للكودينج

Question #2: (9.5 Points)

Given that the academic-staff salaries are stored in "Salaries.csv" file, which is found on the web under:

my_link = "http://rcs.bu.edu/examples/python/data_analysis/Salaries.csv"

```
df.groubly ("rank"[Salary"] .max()
```

5) What is the difference between [df.phd], df.phd and df[[phd]]?

لما ؤكون عمود series بترجع واحد وما فيه باسمه رموز او مسافة وبتسمة dot nation	بس امنة لانها series بترجع بتشتغل على اي اسم	بنوصل من خلال قائمة لعدة اعمدة او حتى لو بدنا نخلي الناتج المرجع data frame

6) Using lambda functions, duplicate the staff salaries (4 M)?

```
df["Salllary_dupple"] = df["Sallay"] .apply (lambda x:x*2)
```

Question #3: (13 Points)

Pandas-Profiling: Assume you have a PostgreSQL database named **ucas_db**, the **ucas_db** DB is located on "192.168.1.10" server. The **ucas_db** has several tables including the **users_tb** and **students_tbl**. The DB user is "root", the DB password is ucas2026.

Answer about the following questions (case-sensitive):

1) Write the code for importing the PostgreSQL connector and pandas-profiling and SQLAlchemy? (__/3M)

```
import psycopg2
import pandas as pd
from ydata_profiling import ProfileReport
```

2) Maintain the basic code reuse principles, write the code necessary to establish the DB engine, connection and create a Dataframe **df** from the sql select statement? (__/11 M)

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(1) The file looks like:

```
rank;discipline;phd;service;sex;salary
AssocProf;B;12;8;Male;119800
Prof;B;28;23;Male;126933
Prof;B;45;45;Male;146856
Prof;A;20;8;Male;102000
AsstProf;B;4;3;Male;91000
Prof;B;18;18;Female;129000
Prof;A;39;36;Female;137000
AssocProf;A;13;8;Female;74830
AsstProf;B;4;2;Female;80225
AsstProf;B;5;0;Female;77000
Prof;B;23;19;Female;151768
Prof;B;25;25;Female;140096
AsstProf;B;11;3;Female;74692
AssocProf;B;11;11;Female;103613
Prof;B;17;17;Female;
```

	rank	discipline	phd	service	sex	salary
0	AssocProf	B	12	8	Male	119800.0
1	Prof	B	28	23	Male	126933.0
2	Prof	B	45	45	Male	146856.0
3	Prof	A	20	8	Male	102000.0
4	AsstProf	B	4	3	Male	91000.0
5	Prof	B	18	18	Female	129000.0
6	Prof	A	39	36	Female	137000.0
7	AssocProf	A	13	8	Female	74830.0
8	AsstProf	B	4	2	Female	80225.0
9	AsstProf	B	5	0	Female	77000.0
10	Prof	B	23	19	Female	151768.0
11	Prof	B	25	25	Female	140096.0
12	AsstProf	B	11	3	Female	74692.0
13	AssocProf	B	11	11	Female	103613.0
14	Prof	B	17	17	Female	NaN

Answer the following questions:

1) Write the code necessary to load the file into a DataFrame df?

```
import pandas as pd
link ="..."
df= pd.read_csv(link)
```

2) Write the code necessary to create a subset DataFrame for the column rank?

```
rank_= df.rank()
```

3) Make a copy from the df above, then write the steps necessary to drop Missing values from the DataFrame?

```
df_copy= df.copy()
df_without_NAN =
df_copy.dropna()
```

4) In one line of code, find the max salary for each professor rank?

```
df.groubly ("rank"[Salary"] .max()
```

5) What is the difference between [df.phd], df.phd and df[[phd]]?

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```
DB_USER = "root"
DB_PASSWORD = "sarah__"
DB_HOST = "192.168.1.10"
DB_PORT = "5432"
DB_NAME = "BDA EX"
```

```
from sqlalchemy import create_engine
URL =
f"postgresql://{DB_USER}:{DB_PASSWORD}@{DB_HOST}:{DB_PORT}/{DB_NAME}"
engine = create_engine(URL)
```

```
query = "SELECT * FROM students_tbl"
df = pd.read_sql_query(query, engine)
```

3) Maintain the basic code reuse principles, write the code necessary to create a profiling report for the **Students_tbl**, you should save the report as HTML file "PTUKStudentsReport" file and the Report should have a title "PTUK Students Report 2023" (__/6M)

```
profile = ProfileReport(df,title="PTUK Students
Report 2023")
profile.to_file("PTUKStudentsReport.html")
```

END OF QUESTIONS, GOOD LUCK :)